What Changed in the Reframe

A note for co-authors on the cascaded oriented-detection manuscript

27 August 2026

The short version

The previous draft was a survey of our own design space. It is now an argument about other people's results. The experiments are largely the same; what changed is what they are used to claim, and four numbers that turned out to be wrong.

The old argument

When Does Tile Gating Earn Its Complexity? An Empirical Study for Oriented Detection in Remote Sensing, with four stated contributions: a Pareto sweep over gate backbone $\times$ calibration $\times$ class; an empirical "sparsity ceiling" that closely tracked the empty-tile rate to within ± 5 percentage points on four data points; an orthogonality result showing that gating compounds with detector size; and two negative co-design results, namely that detector-to-gate distillation and joint shared-backbone training both fail.

Its logic was that the architecture is not novel, so the contribution is measuring it carefully. That placed us in the same category as OAN (Xie et al., *Sci. China Inf. Sci.* 2023) and R^2-CNN (Pang et al., *IEEE TGRS* 2019), doing a similar thing more thoroughly. Thoroughness is not a defence against a paper that already did the thing, which is why the positioning was weak.

The new argument

Rethinking Reported Speedups in Cascaded Oriented Object Detection. The claim is that the speedup any tile gate can achieve is set by the data rather than by the gate, and that this accounts for the range of speedups already published. Six things changed substantively.

The ceiling became an identity rather than a correlation. Previously savings "closely tracked" $1 - p_+$ within ± 5 pp across four points. The per-tile cost is now written exactly, and the accept rate factors through the positive-tile rate and the gate's true and false positive rates. The ± 5 pp does not shrink, it disappears, because what was residual error is now three named terms. The identity reproduces all 3990 measured operating points to floating-point precision.

The ceiling stopped being a ceiling. The old draft reported savings above its own ceiling, 87.1 % against 82.6 % on ships, and attributed the excess to tile overlap. The excess is exactly the compute saved by rejecting tiles that did contain objects. The headline therefore changed from "we save 87 %" to "82 % is free and the remainder is paid for in missed detections". On HRSC2016, 94 % of an apparent saving is discarded detections. This finding was previously hidden inside a number we were pleased with.

Competing papers became supporting evidence. OAN's own patch-removal ablation traces our envelope on their data, holding recall above 99.7 % up to 30 % removal and eroding past it, which is where the empty-patch rate we measure independently sits. Their headline speedup is what a gating task with $p_+ = 0.68$ permits. Three published systems, skipping 35 %, 65 % and about 99 % of tiles respectively, each sit at their own $1 - p_+$.

A cost dimension we had suppressed became a section. Our own wall-clock measurements contradicted the GFLOPs story since May and the draft simply never cited them. The gate is 0.33 % of the detector in arithmetic and 27.6 % in single-tile latency, a factor of 84. Re-priced in latency, the HRSC2016 saving becomes negative.

One contribution was retracted. Reimplementing OAN's actual objectness head shows that joint training works: it costs nothing in detection accuracy, produces a better gate than training one separately, and removes the overhead term entirely. Our earlier negative result reflected the head we built, not co-design. The distillation negative result stands.

Evidence grew from four points to nineteen gating tasks spanning empty-tile rates from 2.4 % to 97.9 %, built from DOTA-1.5 classes, a sixteen-class union, and the HRSC2016 transfer.

Numbers that changed, and why

Quantity	Was	Now, and the reason
mAP tolerance	stated 3 pp, applied 97 % relative	3 pp throughout. Ships 87.1 to 88.2 %, planes 83.2 to 85.7 %, small-vehicle 69.0 to 76.0 %
Full-pass baseline	each class charged for a different arbitrary gate	974,780 GFLOPs for all three. The old value depended on filesystem ordering
HRSC2016 positive rate	94 %, hardcoded	97.9 % from the labels. HRSC was also absent from a figure whose caption claimed four datasets
Object loss at $\tau = 0.5$	planes 6.7 %, small-vehicle 7.4 %	11.9 % for both. The old figures used gates appearing nowhere else in the paper

What survived unchanged

Classical score calibration reduces expected calibration error substantially and does not move the cascade frontier, for the mechanical reason that a threshold sweep already extracts everything the score ordering contains. The distinction between calibrating probabilities and selecting an operating point stands. Detector-to-gate distillation fails, for the reason previously identified. Sensor source dominates the stratified picture, with mAP varying roughly fiftyfold across sources on ships. Gating compounds with detector size.

Status

Ten pages, fifteen sections, six figures, two tables, sixty-nine references, all cited and all verified against the published record. Every number in the prose is generated from the results directory rather than typed, so a stale value becomes a build-time change rather than a silent inconsistency.

The one experiment still worth running before submission is a seed and λ sweep on the fused-gate comparison. Its advantage in compute saved is 2.3 pp, which is within plausible seed variation, so the manuscript claims the qualitative result and explicitly declines to claim the margin.